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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: **M.Tech**
Name of the Paper : **Machine Learning**

Semester: **1st**
Paper Code: **MCS 128**

Time: 3 Hours

Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q 1) (2X10=20 Marks) (CO2, CO4, CO5)

(a) Explain the differences between underfitting and overfitting in machine learning. How do techniques like regularization and dropout address these issues?

(b) Describe the Elbow Method and Silhouette Scores for determining the optimal number of clusters in K-means. Illustrate their application using the dataset:

Dataset: { (1, 2), (3, 4), (10, 12), (8, 10), (4, 5), (11, 13) }.

(c) What is the concept of Maximum A Posteriori Estimation (MAP)? Compare and contrast it with Maximum Likelihood Estimation (MLE).

Q 2) (2X10=20 Marks) (CO2, CO4, CO5, CO6)

(a) Consider the following dataset:

A	B	Class
1	0	Positive
0	0	Negative
0	1	Negative
0	1	Negative
0	0	Positive
1	0	Positive

Estimate the parameters for training a Naïve Bayes classifier and classify the test sample A = 0, B = 1 using the estimated parameters.

(b) Discuss Principal Component Analysis (PCA). Perform PCA on the following 2D dataset and find the principal components:

Dataset: { (2, 4), (3, 3), (4, 2), (5, 5) }.

(c) What are autoencoders? Describe their structure and provide an example of their real-world application in anomaly detection.

Q 3) (2X10=20 Marks) (CO2, CO4, CO5, CO6)

(a) Explain the forward pass of a convolution neural network with the following configuration:

Input size: (64 X 64 X 3)

- Layer 1: 8 filters of size (3 X 3), Stride = 1, Padding = 1
- Max Pooling: Filter size (2 X 2), Stride = 2
- Layer 2: 16 filters of size (3 X 3), Stride = 1, Padding = 1
- Fully Connected Layer: Output size = 100
- Softmax Layer for classification with 5 categories.

(b) Discuss the concept of backpropagation in training neural networks. Illustrate weight updates using gradient descent for a simple two-layer neural network.

(c) What are the applications of deep learning? Provide examples of its use in healthcare, autonomous vehicles, and natural language processing.

Q 4) (2X10=20 Marks) (CO1, CO2)

(a) Compare and contrast hierarchical clustering with K-means clustering. Discuss the advantages of hierarchical clustering using a dataset with the following points:

Dataset: { (1, 2), (3, 4), (5, 6), (8, 8) }.

(b) Apply one iteration of the K-means algorithm to the dataset: {(2, 2), (4, 4), (6, 6), (8, 8)} with $k = 2$ and initial centroids at (2, 2) and (6, 6). Update the centroids after the first iteration.

(c) Provide an overview of reinforcement learning. How does it differ from supervised and unsupervised learning paradigms? Give an example of a reinforcement learning application in robotics.

Q 5) (2X10=20 Marks) (CO2, CO5)

(a) What is the bias-variance trade-off? Discuss its impact on a model's performance and generalization ability. Use an example to illustrate the trade-off.

(b)Explore activation functions in neural networks. Provide examples of one non-linear activation function and one linear activation function, and discuss their significance in network performance.

(c) Define a perceptron. Explain how the perceptron algorithm works with the following dataset:

x1	x2	Output
1	1	1
1	0	0
0	1	0
0	0	0

Update the weights for one iteration assuming the initial weights are 0 and the learning rate is 1.